

Technology Stack

Project: **Auto Ticket Classification using Flow Design**

Technology Stack (Architecture & Tools)

S.No	Component	Description	Technology
1	User Interface	Ticket submission and agent dashboard	HTML, CSS, JavaScript / Streamlit
2	Flow Engine	Decision-flow logic for routing and escalation	Python workflow logic / Node-RED / Power Automate
3	Application Logic	Preprocessing and classification service	Python, Flask / FastAPI
4	ML Model	Text classification for category and priority	Scikit-learn (TF-IDF + Logistic Regression / SVM)
5	GenAI (optional)	Summary and zero-shot classification for tough tickets	LLM API
6	Database	Stores tickets, labels and feedback	SQLite / MySQL / MongoDB
7	Notifications	Acknowledgement and priority alerts	SMTP email / Slack webhook
8	Deployment	Hosting the application	Cloud (Render / AWS / Azure)

Characteristics

- **Open-source frameworks:** Python, Flask/FastAPI, Scikit-learn, NLTK/spaCy.
- **Security:** HTTPS, authentication and role-based access control.
- **Scalable architecture:** flow engine, classifier and database run as separate services.
- **Availability:** cloud deployment with automatic restarts.